



SPINGUARD PRO

Advanced Ring Spinning Frame Control & Protection Software



KEY FEATURES

1 Smart Doffing Control (SDC)

Automatically manages the ring rail position throughout the doffing sequence. Upon completion of doffing, the ring rail is raised to a predefined height and automatically returns to its normal operating position once the main motor reaches the required running condition.

1

The system provides two operating modes:

(1) Controlled / Optimized Speed Doffing, and (2) Running / Momentum Speed Doffing

This flexible control allows the doffing sequence to be adapted to different machine operating conditions, while ensuring smooth ring rail movement and minimizing yarn end-downs, yarn tension disturbances, and yarn breakage during machine restart.

2 Power Failure Recovery (PFR)

Provides a controlled ring rail recovery sequence following an unexpected power interruption. Upon restoration of power, the rail is automatically moved to a predefined recovery position and, after a certain delay, returns to its actual operating position. The controlled recovery prevents yarn tension disturbances during restart, with sudden rail movement. This function helps minimize yarn breakage and supports a stable and reliable machine recovery after power interruptions.

2

3 Doffing Switches Supervision (DSS)

Provides software-based protection during the doffing sequence by continuously monitoring the ring rail limit-switch feedback and validating the expected operating conditions. If the required limit-switch signal is not received or an abnormal condition is detected, the control system restricts further rail movement and maintains the rail within a predefined safe position. This additional protection helps prevent rail over-travel, mechanical interference, and potential damage to machine components in the event of a limit-switch or feedback failure.

3

4 Full Doff Verification (FDV)

Ensures completion of the complete doffing cycle by continuously monitoring the predefined yarn length. The system prevents premature termination of the doffing sequence and ensures that the required doffing position is achieved before allowing the cycle to proceed. This function improves doffing consistency, reduces incomplete doffing events, and helps maintain reliable machine operation across repeated production cycles.

4

5 Overhead Cleaner Interlock (OHCI)

Automatically manages the position and operating status of the overhead cleaner during the doffing operation. At the beginning of the doffing sequence, the cleaner is commanded to stop at the designated machine-head position to prevent interference with operators, doffing equipment, and moving machine components. The control logic maintains the cleaner in the required position throughout the critical doffing period and permits normal operation once the doffing sequence is safely completed.

5

6 Automatic Lubrication Control (ALC)

Controls the electrically operated lubrication oil pump according to predefined operating conditions, timing parameters, and machine requirements. The control logic manages pump operation to provide lubrication only when required, preventing unnecessary continuous operation or excessive oil delivery. This optimized control helps reduce over-lubrication, minimize oil consumption, improve lubrication consistency, and extend the service life of the pump and associated lubrication-system components.

6

Why choose SPINGUARD PRO?

- ✓ Real-time Machine Monitoring
- ✓ Flexible & Configurable Parameters
- ✓ Simplified Fault Diagnosis
- ✓ Reduced Yarn End-Downs After Restart
- ✓ Reduced Downtime After Power Failure
- ✓ Monitored No. of Doff & Machine Downtime
- ✓ Optimized Doffing Cycles with Flexible Modes
- ✓ Short Length Doffing Control
- ✓ Consistent Machine Performance
- ✓ Enhanced Machine Protection in Abnormal Ring Rail Operations
- ✓ Reduced Equipment Wear with Controlled Lubrication
- ✓ Energy Conservation in Maintenance Mode